

8.1 Distributions

Definition 54 (Cumulative distribution function). For a univariate distribution $p(x)$, the CDF is defined as

$$cdf(y) \equiv p(x \leq y) = \langle \mathbb{I}[x \leq y] \rangle_{p(x)} \quad (8.1.1)$$

For an unbounded domain, $cdf(-\infty) = 0$ and $cdf(\infty) = 1$.

8.2 Summarising distributions

Definition 55 (Mode). The mode x_* of a distribution $p(x)$ is the state of x at which the distribution takes its highest value, $x_* = \underset{x}{\operatorname{argmax}} p(x)$. A distribution could have more than one mode (be multi-modal). A widespread abuse of terminology is to refer to any isolated local maximum of $p(x)$ to be a mode.

Definition 56 (Averages and Expectation).

$$\langle f(x) \rangle_{p(x)} \quad (8.2.1)$$

denotes the average or expectation of $f(x)$ with respect to the distribution $p(x)$. A common alternative notation is

$$\mathbb{E}(x) \quad (8.2.2)$$

When the context is clear, one may drop the notational dependency on $p(x)$. The notation

$$\langle f(x) | y \rangle \quad (8.2.3)$$

is shorthand for the average of $f(x)$ conditioned on knowing the state of variable y , *i.e.* the average of $f(x)$ with respect to the distribution $p(x|y)$.

An advantage of the expectation notations is that they hold whether the distribution is over continuous or discrete variables. In the discrete case

$$\langle f(x) \rangle \equiv \sum_x f(x) p(x) \quad (8.2.4)$$

and for continuous variables,

$$\langle f(x) \rangle \equiv \int_{-\infty}^{\infty} f(x) p(x) dx \quad (8.2.5)$$

The reader might wonder what $\langle x \rangle$ means when x is discrete. For example, if $\text{dom}(x) = \{\text{apple}, \text{orange}, \text{pear}\}$, with associated probabilities $p(x)$ for each of the states, what does $\langle x \rangle$ refer to? Clearly, $\langle f(x) \rangle$ makes sense if $f(x = x)$ maps the state x to a numerical value. For example $f(x = \text{apple}) = 1$, $f(x = \text{orange}) = 2$, $f(x = \text{pear}) = 3$ for which $\langle f(x) \rangle$ is meaningful. Unless the states of the discrete variable are associated with a numerical value, then $\langle x \rangle$ has no meaning.

Definition 57 (Moments). The k^{th} moment of a distribution is given by the average of x^k under the distribution:

$$\langle x^k \rangle_{p(x)} \quad (8.2.6)$$

For $k = 1$, we have the mean, typically denoted by μ ,

$$\mu \equiv \langle x \rangle \quad (8.2.7)$$

Definition 58 (Variance and Correlation).

$$\sigma^2 \equiv \langle (x - \langle x \rangle)^2 \rangle_{p(x)} \quad (8.2.8)$$

The square root of the variance, σ is called the *standard deviation*. The notation $\text{var}(x)$ is also used to emphasise for which variable the variance is computed. The reader may show that an equivalent expression is

$$\sigma^2 \equiv \langle x^2 \rangle - \langle x \rangle^2 \quad (8.2.9)$$

For a multivariate distribution the matrix with elements

$$\Sigma_{ij} = \langle (x_i - \mu_i)(x_j - \mu_j) \rangle \quad (8.2.10)$$

where $\mu_i = \langle x_i \rangle$ is called the *covariance matrix*. The diagonal entries of the covariance matrix contain the variance of each variable. An equivalent expression is

$$\Sigma_{ij} = \langle x_i x_j \rangle - \langle x_i \rangle \langle x_j \rangle \quad (8.2.11)$$

The *correlation matrix* has elements

$$\rho_{ij} = \left\langle \frac{(x_i - \mu_i)}{\sigma_i} \frac{(x_j - \mu_j)}{\sigma_j} \right\rangle \quad (8.2.12)$$

where σ_i is the deviation of variable x_i . The correlation is a normalised form of the covariance so that each element is bounded $-1 \leq \rho_{ij} \leq 1$.

For independent variables x_i and x_j , $x_i \perp\!\!\!\perp x_j \mid \emptyset$ the covariance Σ_{ij} is zero. Similarly independent variables have zero correlation – they are ‘uncorrelated’. Note however that the converse is not generally true – two variables can be uncorrelated but dependent. A special case is for when x_i and x_j are Gaussian distributed then independence is equivalent to being uncorrelated, see exercise(81).

Definition 59 (Skewness and Kurtosis). The skewness is a measure of the asymmetry of a distribution:

$$\gamma_1 \equiv \frac{\left\langle (x - \langle x \rangle)^3 \right\rangle_{p(x)}}{\sigma^3} \quad (8.2.13)$$

where σ^2 is the variance of x with respect to $p(x)$. A positive skewness means the distribution has a heavy tail to the right. Similarly, a negative skewness means the distribution has a heavy tail to the left.

The kurtosis is a measure of how peaked around the mean a distribution is:

$$\gamma_2 \equiv \frac{\left\langle (x - \langle x \rangle)^4 \right\rangle_{p(x)}}{\sigma^4} - 3 \quad (8.2.14)$$

A distribution with positive kurtosis has more mass around its mean than would a Gaussian with the same mean and variance. These are also called *super Gaussian*. Similarly a negative kurtosis (*sub Gaussian*) distribution has less mass around its mean than the corresponding Gaussian. The kurtosis is defined such that a Gaussian has zero kurtosis (which accounts for the -3 term in the definition).

Definition 60 (Empirical Distribution). For a set of datapoints $x^1, \dots, x^N$, which are states of a random variable x , the empirical distribution has probability mass distributed evenly over the datapoints, and zero elsewhere.

For a discrete variable x the empirical distribution is

$$p(x) = \frac{1}{N} \sum_{n=1}^N \mathbb{I}[x = x^n] \quad (8.2.15)$$

where N is the number of datapoints.

For a continuous distribution we have

$$p(x) = \frac{1}{N} \sum_{n=1}^N \delta(x - x^n) \quad (8.2.16)$$

where $\delta(x)$ is the Dirac Delta function.

The sample mean of the datapoints is given by the

$$\hat{\mu} = \frac{1}{N} \sum_{n=1}^N x^n \quad (8.2.17)$$

and the sample variance is given by the

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N (x^n - \hat{\mu})^2 \quad (8.2.18)$$

For vectors the sample mean vector has elements

$$\hat{\mu}_i = \frac{1}{N} \sum_{n=1}^N x_i^n \quad (8.2.19)$$

and sample covariance matrix has elements

$$\hat{\Sigma}_{ij} = \frac{1}{N} \sum_{n=1}^N (x_i^n - \hat{\mu}_i) (x_j^n - \hat{\mu}_j) \quad (8.2.20)$$

Definition 61 (Delta function). For continuous x , we define the Dirac delta function

$$\delta(x - x_0) \quad (8.2.21)$$

which is zero everywhere except at x_0 , where there is a spike. $\int_{-\infty}^{\infty} \delta(x - x_0) dx = 1$ and

$$\int_{-\infty}^{\infty} \delta(x - x_0) f(x) dx = f(x_0) \quad (8.2.22)$$

One can view the Dirac delta function as an infinitely narrow Gaussian: $\delta(x - x_0) = \lim_{\sigma \rightarrow 0} \mathcal{N}(x|x_0, \sigma^2)$.

The *Kronecker delta*,

$$\delta_{x,x_0} \quad (8.2.23)$$

is similarly zero everywhere, except for $\delta_{x_0,x_0} = 1$. The Kronecker delta is equivalent to $\delta_{x,x_0} = \mathbb{I}[x = x_0]$. We use the expression $\delta(x, x_0)$ to denote either the Dirac or Kronecker delta, depending on the context.

8.2.1 Estimator bias

Definition 62 (Unbiased estimator). Given data $\mathcal{X} = x^1, \dots, x^N$, from a distribution $p(x|\theta)$ we can use the data $\mathcal{X}$ to estimate the parameter θ that was used to generate the data. The estimator is a function of the data, which we write $\hat{\theta}(\mathcal{X})$. For an *unbiased estimator*

$$\langle \hat{\theta}(\mathcal{X}) \rangle_{p(\mathcal{X}|\theta)} = \theta \quad (8.2.24)$$

More generally, one can consider any estimating function $\hat{\psi}(\mathcal{X})$ of data. This is an unbiased estimator of a quantity ψ if $\langle \hat{\psi}(\mathcal{X}) \rangle_{p(\mathcal{X})} = \psi$.



Figure 8.1: Empirical distribution over a discrete variable with 4 states. The empirical samples consist of n samples at each of states 1, 2, 4 and $2n$ samples at state 3 where $n > 0$. On normalising this gives a distribution with values 0.2, 0.2, 0.4, 0.2 over the 4 states.

A classical example for estimator bias are those of the mean and variance. Let

$$\hat{\mu}(\mathcal{X}) = \frac{1}{N} \sum_{n=1}^N x^n \quad (8.2.25)$$

This is an unbiased estimator of the mean $\langle x \rangle_{p(x)}$ since

$$\langle \hat{\mu}(\mathcal{X}) \rangle_{p(x)} = \frac{1}{N} \sum_{n=1}^N \langle x^n \rangle_{p(x)} = \frac{1}{N} N \langle x \rangle_{p(x)} = \langle x \rangle_{p(x)} \quad (8.2.26)$$

On the other hand, consider the estimator of the variance,

$$\hat{\sigma}^2(\mathcal{X}) = \frac{1}{N} \sum_{n=1}^N (x^n - \hat{\mu}(\mathcal{X}))^2 \quad (8.2.27)$$

This is biased since (omitting a few lines of algebra)

$$\langle \hat{\sigma}^2(\mathcal{X}) \rangle_{p(x)} = \frac{1}{N} \sum_{n=1}^N \langle (x^n - \hat{\mu}(\mathcal{X}))^2 \rangle = \frac{N-1}{N} \sigma^2 \quad (8.2.28)$$

8.3 Discrete Distributions

Definition 63 (Bernoulli Distribution). The Bernoulli distribution concerns a discrete binary variable x , with $\text{dom}(x) = \{0, 1\}$. The states are not merely symbolic, but real values 0 and 1.

$$p(x = 1) = \theta \quad (8.3.1)$$

From normalisation, it follows that $p(x = 0) = 1 - \theta$. From this

$$\langle x \rangle = 0 \times p(x = 0) + 1 \times p(x = 1) = \theta \quad (8.3.2)$$

The variance is given by $\text{var}(x) = \theta(1 - \theta)$.

Definition 64 (Categorical Distribution). The categorical distribution generalises the Bernoulli distribution to more than two (symbolic) states. For a discrete variable x , with symbolic states $\text{dom}(x) = \{1, \dots, C\}$,

$$p(x = c) = \theta_c, \quad \sum_c \theta_c = 1 \quad (8.3.3)$$

The Dirichlet is conjugate to the categorical distribution.

Definition 65 (Binomial Distribution). The Binomial describes the distribution of a discrete two-state variable x , with $\text{dom}(x) = \{1, 0\}$ where the states are symbolic. The probability that in n Bernoulli Trials (independent samples), k ‘success’ states 1 will be observed is

$$p(y = k | \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k} \quad (8.3.4)$$

where $\binom{n}{k}$ is the binomial coefficient. The mean and variance are

$$\langle y \rangle = n\theta, \quad \text{var}(x) = n\theta(1 - \theta) \quad (8.3.5)$$

The Beta distribution is the conjugate prior for the Binomial distribution.

Definition 66 (Multinomial Distribution). Consider a multi-state variable x , with $\text{dom}(x) = \{1, \dots, K\}$, with corresponding state probabilities $\theta_1, \dots, \theta_K$. We then draw n samples from this distribution. The probability of observing the state 1 y_1 times, state 2 y_2 times, $\dots$, state K y_K times in the n samples is

$$p(y|\theta) = \frac{n!}{y_1! \dots y_K!} \prod_{i=1}^n \theta_i^{y_i} \quad (8.3.6)$$

where $n = \sum_{i=1}^n y_i$.

$$\langle y_i \rangle = n\theta_i, \quad \text{var}(y_i) = n\theta_i(1 - \theta_i), \quad \langle y_i y_j \rangle - \langle y_i \rangle \langle y_j \rangle = -n\theta_i \theta_j \quad (i \neq j) \quad (8.3.7)$$

The Dirichlet distribution is the conjugate prior for the multinomial distribution.

Definition 67 (Poisson Distribution). The Poisson distribution can be used to model situations in which the expected number of events scales with the length of the interval within which the events can occur. If λ is the expected number of events per unit interval, then the distribution of the number of events x within an interval $t\lambda$ is

$$p(x = k|\lambda) = \frac{1}{k!} e^{-\lambda t} (\lambda t)^k, \quad k = 0, 1, 2, \dots \quad (8.3.8)$$

For a unit length interval ($t = 1$),

$$\langle x \rangle = \lambda, \quad \text{var}(x) = \lambda \quad (8.3.9)$$

The Poisson distribution can be derived as a limiting case of a Binomial distribution in which the success probability scales as $\theta = \lambda/n$, in the limit $n \rightarrow \infty$.

8.4 Continuous Distributions

8.4.1 Bounded distributions

Definition 68 (Uniform distribution). For a variable x , the distribution is uniform if $p(x) = \text{const.}$ over the domain of the variable.

Definition 69 (Exponential Distribution). For $x \geq 0$,

$$p(x|\lambda) \equiv \lambda e^{-\lambda x} \quad (8.4.1)$$

One can show that for rate λ

$$\langle x \rangle = \frac{1}{\lambda}, \quad \text{var}(x) = \frac{1}{\lambda^2} \quad (8.4.2)$$

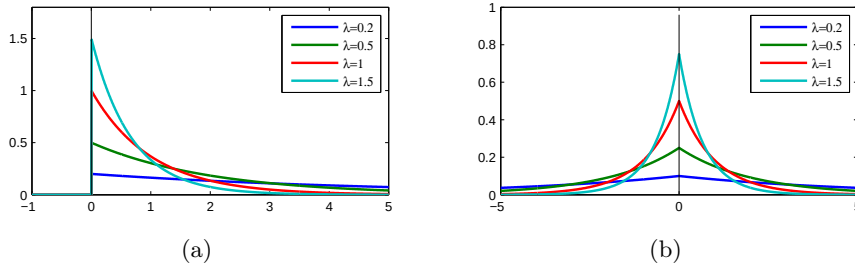


Figure 8.2: **(a)**: Exponential distribution. **(b)**: Laplace (double exponential) distribution.

The alternative parameterisation $b = 1/\lambda$ is called the scale.

Definition 70 (Gamma Distribution).

$$Gam(x|\alpha, \beta) = \frac{1}{\beta\Gamma(\alpha)} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\frac{x}{\beta}}, \quad x \geq 0, \alpha > 0, \beta > 0 \quad (8.4.3)$$

α is called the shape parameter, β is the scale parameter and

$$\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt \quad (8.4.4)$$

The parameters are related to the mean and variance through

$$\alpha = \left(\frac{\mu}{s}\right)^2, \quad \beta = \frac{s^2}{\mu} \quad (8.4.5)$$

where μ is the mean of the distribution and s is the standard deviation. The mode is given by $(\alpha - 1)\beta$, for $\alpha \geq 1$.

An alternative parameterisation uses the inverse scale

$$Gam^{is}(x|\alpha, b) = Gam(x|\alpha, 1/b) \propto x^{\alpha-1} e^{-bx} \quad (8.4.6)$$

Definition 71 (Inverse Gamma distribution).

$$InvGam(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{1}{x^{\alpha+1}} e^{-\beta/x} \quad (8.4.7)$$

This has mean $\beta/(\alpha - 1)$ for $\alpha > 1$ and variance $\frac{\beta^2}{(\alpha-1)^2(\alpha-2)}$ for $\alpha > 2$.

Definition 72 (Beta Distribution).

$$p(x|\alpha, \beta) = B(x|\alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1 \quad (8.4.8)$$

where the beta function is defined as

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)} \quad (8.4.9)$$

and $\Gamma(x)$ is the gamma function. Note that the distribution can be flipped by interchanging x for $1 - x$, which is equivalent to interchanging α and β .

The mean is given by

$$\langle x \rangle = \frac{\alpha}{\alpha + \beta}, \quad \text{var}(x) = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)} \quad (8.4.10)$$

8.4.2 Unbounded distributions

Definition 73 (Laplace (Double Exponential) Distribution).

$$p(x|\lambda) \equiv \lambda e^{-\frac{1}{b}|x-\mu|} \quad (8.4.11)$$

For scale b

$$\langle x \rangle = \mu, \quad \text{var}(x) = 2b^2 \quad (8.4.12)$$

Univariate Gaussian distribution

The Gaussian distribution is an important distribution in science. It's technical description is given in definition(74).

Definition 74 (Univariate Gaussian Distribution).

$$p(x|\mu, \sigma^2) = \mathcal{N}(x|\mu, \sigma^2) \equiv \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad (8.4.13)$$

where μ is the mean of the distribution, and σ^2 the variance. This is also called the *normal distribution*.

One can show that the parameters indeed correspond to

$$\mu = \langle x \rangle_{\mathcal{N}(x|\mu, \sigma^2)}, \quad \sigma^2 = \langle (x - \mu)^2 \rangle_{\mathcal{N}(x|\mu, \sigma^2)} \quad (8.4.14)$$

For $\mu = 0$ and $\sigma = 1$, the Gaussian is called the *standard normal distribution*.

Definition 75 (Student's t -distribution).

$$p(x|\mu, \lambda, \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \left(\frac{\lambda}{\nu\pi} \right)^{\frac{1}{2}} \left[1 + \frac{\lambda(x - \mu)^2}{\nu} \right]^{-\frac{\nu+1}{2}} \quad (8.4.15)$$

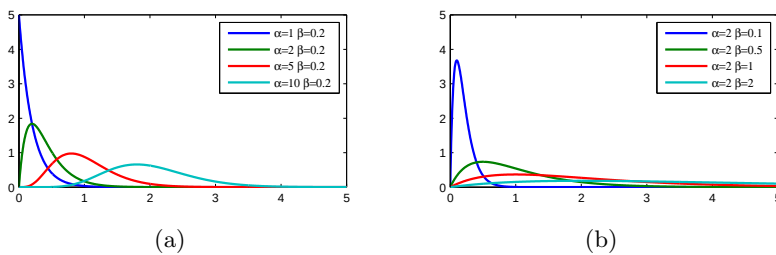


Figure 8.3: **(a)**: Gamma distribution with varying β for fixed α . **(b)**: Gamma distribution with varying α for fixed β .

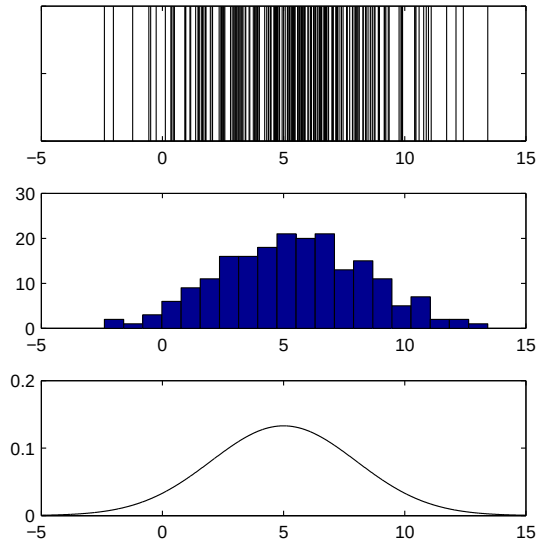


Figure 8.4: Top: 200 datapoints $x^1, \dots, x^{200}$ drawn from a Gaussian distribution. Each vertical line denotes a datapoint at the corresponding x value on the horizontal axis. Middle: Histogram using 10 equally spaced bins of the datapoints. Bottom: Gaussian distribution $\mathcal{N}(x|\mu=5, \sigma=3)$ from which the datapoints were drawn. In the limit of an infinite amount of data, and limitingly small bin size, the normalised histogram tends to the Gaussian probability density function.

where μ is the mean, ν the degrees of freedom, and λ scales the distribution. The variance is given by

$$\text{var}(x) = \frac{\nu}{\lambda(\nu-2)}, \text{ for } \nu > 2 \quad (8.4.16)$$

For $\nu \rightarrow \infty$ the distribution tends to a Gaussian with mean μ and variance $1/\lambda$. As ν decreases the tails of the distribution become fatter.

The t -distribution can be derived from a *scaled mixture*

$$p(x|a, b) = \int_{\tau=0}^{\infty} \mathcal{N}(x|\mu, \tau^{-1}) \text{Gam}^{is}(\tau|a, b) d\tau \quad (8.4.17)$$

$$= \left(\frac{\tau}{2\pi}\right)^{\frac{1}{2}} \int_{\tau=0}^{\infty} e^{-\frac{\tau}{2}(x-\mu)^2} b^a e^{-b\tau} \tau^{a-1} \frac{1}{\Gamma(a)} d\tau \quad (8.4.18)$$

$$= \frac{b^a}{\Gamma(a)} \frac{\Gamma(a + \frac{1}{2})}{\sqrt{2\pi}} \frac{1}{\left(b + \frac{1}{2}(x-\mu)^2\right)^{a+\frac{1}{2}}} \quad (8.4.19)$$

It is conventional to reparameterise using $\nu = 2a$ and $\lambda = a/b$.

8.5 Multivariate Distributions

Definition 76 (Dirichlet Distribution). The Dirichlet distribution is a distribution on probability distributions:

$$p(\alpha) = \frac{1}{Z(\mathbf{u})} \delta\left(\sum_{i=1}^Q \alpha_i - 1\right) \prod_{q=1}^Q \alpha_q^{u_q-1} \quad (8.5.1)$$

where

$$Z(\mathbf{u}) = \frac{\prod_{q=1}^Q \Gamma(u_q)}{\Gamma\left(\sum_{q=1}^Q u_q\right)} \quad (8.5.2)$$

It is conventional to denote the distribution as

$$\text{Dirichlet}(\alpha|\mathbf{u}) \quad (8.5.3)$$

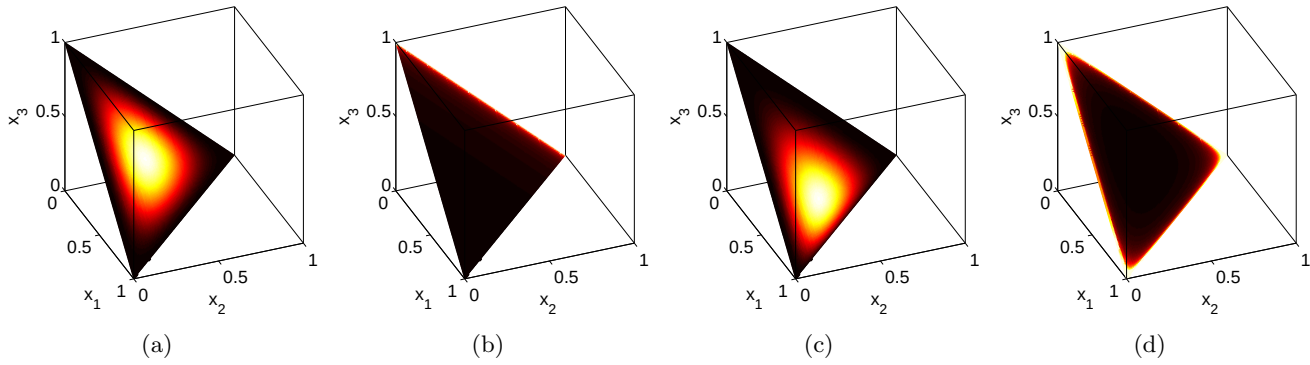


Figure 8.5: Dirichlet distribution with parameter (u_1, u_2, u_3) displayed on the simplex $x_1, x_2, x_3 \geq 0, x_1 + x_2 + x_3 = 1$. Black denotes low probability and white high probability. **(a)**: $(3, 3, 3)$ **(b)**: $(0.1, 1, 1)$. **(c)**: $(4, 3, 2)$. **(d)**: $(0.05, 0.05, 0.05)$.

The parameter $\mathbf{u}$ controls how strongly the mass of the distribution is pushed to the corners of the simplex. Setting $u_q = 1$ for all q corresponds to a uniform distribution, fig(8.5). In the binary case $Q = 2$, this is equivalent to a Beta distribution.

The product of two Dirichlet distributions is

$$\text{Dirichlet}(\theta|\mathbf{u}_1) \text{Dirichlet}(\theta|\mathbf{u}_2) = \text{Dirichlet}(\theta|\mathbf{u}_1 + \mathbf{u}_2) \quad (8.5.4)$$

Marginal of a Dirichlet:

$$\int_{\theta_j} \text{Dirichlet}(\theta|\mathbf{u}) = \text{Dirichlet}(\theta_{\setminus j}|\mathbf{u}_{\setminus j}) \quad (8.5.5)$$

The marginal of a single component θ_i is a Beta distribution:

$$p(\theta_i) = B\left(\theta_i|u_i, \sum_{j \neq i} u_j\right) \quad (8.5.6)$$

8.6 Multivariate Gaussian

The multivariate Gaussian plays a central role throughout this book and as such we discuss its properties in some detail.

Definition 77 (Multivariate Gaussian Distribution).

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \equiv \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})} \quad (8.6.1)$$

where $\boldsymbol{\mu}$ is the mean vector of the distribution, and $\boldsymbol{\Sigma}$ the covariance matrix. The inverse covariance $\boldsymbol{\Sigma}^{-1}$ is called the *precision*.

One may show

$$\boldsymbol{\mu} = \langle \mathbf{x} \rangle_{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})}, \quad \boldsymbol{\Sigma} = \left\langle (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \right\rangle_{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})} \quad (8.6.2)$$

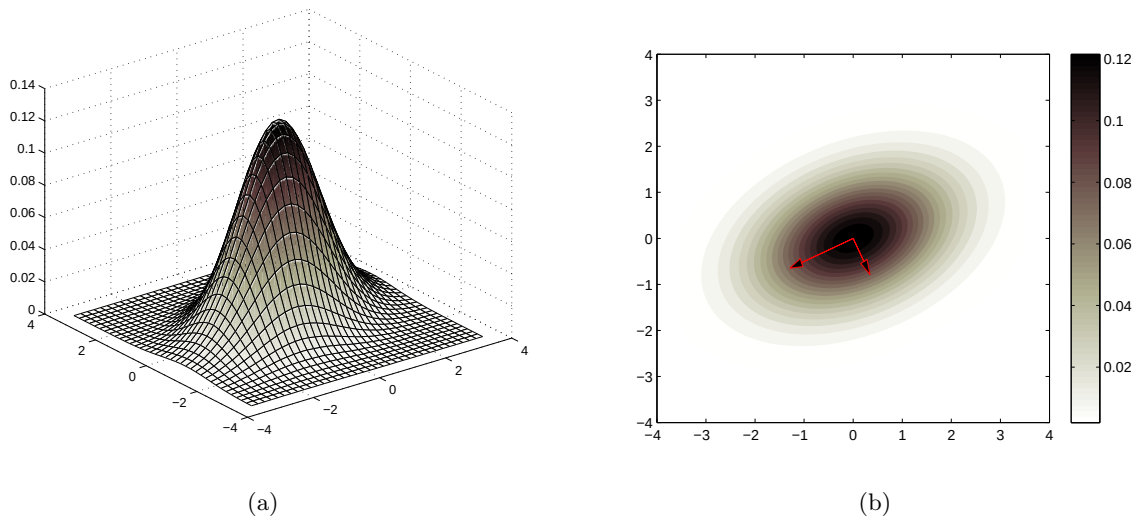


Figure 8.6: **(a)**: Bivariate Gaussian with mean $(0,0)$ and covariance $[1, 0.5; 0.5, 1.75]$. Plotted on the vertical axis is the probability density value $p(x)$. **(b)**: Probability density contours for the same bivariate Gaussian. Plotted are the unit eigenvectors scaled by the square root of their eigenvalues, $\sqrt{\lambda_i}$.

The multivariate Gaussian is given in definition(77). Note that $\det(\rho \mathbf{M}) = \rho^D \det(\mathbf{M})$, where $\mathbf{M}$ is a $D \times D$ matrix, which explains the dimension independent notation in the normalisation constant of definition(77).

The *moment representation* uses $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ to parameterise the Gaussian. The alternative *canonical representation*

$$p(\mathbf{x}|\mathbf{b}, \mathbf{M}, c) = ce^{-\frac{1}{2}\mathbf{x}^\top \mathbf{M} \mathbf{x} + \mathbf{x}^\top \mathbf{b}} \quad (8.6.3)$$

is related to the moment representation via

$$\boldsymbol{\Sigma} = \mathbf{M}^{-1}, \quad \boldsymbol{\mu} = \mathbf{M}^{-1}\mathbf{b}, \quad \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} = ce^{\frac{1}{2}\mathbf{b}^\top \mathbf{M}^{-1}\mathbf{b}} \quad (8.6.4)$$

The multivariate Gaussian is widely used and it is instructive to understand the geometric picture. This can be obtained by view the distribution in a different co-ordinate system. First we use that every real symmetric matrix $D \times D$ has an eigen-decomposition

$$\boldsymbol{\Sigma} = \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top \quad (8.6.5)$$

where $\mathbf{E}^\top \mathbf{E} = \mathbf{I}$ and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_D)$. In the case of a covariance matrix, all the eigenvalues λ_i are positive. This means that one can use the transformation

$$\mathbf{y} = \boldsymbol{\Lambda}^{\frac{1}{2}} \mathbf{E}^\top (\mathbf{x} - \boldsymbol{\mu}) \quad (8.6.6)$$

so that

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma} (\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top (\mathbf{x} - \boldsymbol{\mu}) = \mathbf{y}^\top \mathbf{y} \quad (8.6.7)$$

Under this transformation, the multivariate Gaussian reduces to a product of D univariate zero-mean unit variance Gaussians (since the Jacobian of the transformation is a constant). This means that we can view a multivariate Gaussian as a shifted, scaled and rotated version of an isotropic Gaussian in which the centre is given by the mean, the rotation by the eigenvectors, and the scaling by the square root of the eigenvalues, as depicted in fig(8.6b).

Isotropic means ‘same under rotation’. For any isotropic distribution, contours of equal probability are spherical around the origin.

Some useful properties of the Gaussian are as follows:

Definition 78 (Partitioned Gaussian). For a distribution $\mathcal{N}(\mathbf{z}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ defined jointly over two vectors $\mathbf{x}$ and $\mathbf{y}$ of potentially differing dimensions,

$$\mathbf{z} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \quad (8.6.8)$$

with corresponding mean and partitioned covariance

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{pmatrix} \quad (8.6.9)$$

where $\boldsymbol{\Sigma}_{yx} \equiv \boldsymbol{\Sigma}_{xy}^\top$. The marginal distribution is given by

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_{xx}) \quad (8.6.10)$$

and conditional

$$p(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_x + \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}(\mathbf{y} - \boldsymbol{\mu}_y), \boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}) \quad (8.6.11)$$

Definition 79 (Product of two Gaussians). The product of two Gaussians is another Gaussian, with a multiplicative factor, exercise(114):

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \frac{\exp\left(-\frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \mathbf{S}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\right)}{\sqrt{\det(2\pi\mathbf{S})}} \quad (8.6.12)$$

where $\mathbf{S} \equiv \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$ and the mean and covariance are given by

$$\boldsymbol{\mu} = \boldsymbol{\Sigma}_1\mathbf{S}^{-1}\boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_2\mathbf{S}^{-1}\boldsymbol{\mu}_1 \quad \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_1\mathbf{S}^{-1}\boldsymbol{\Sigma}_2 \quad (8.6.13)$$

Definition 80 (Linear Transform of a Gaussian). For the linear transformation

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b} \quad (8.6.14)$$

where $\mathbf{x} \sim \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$\mathbf{y} \sim \mathcal{N}(\mathbf{y}|\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top) \quad (8.6.15)$$

Definition 81 (Entropy of a Gaussian). The differential entropy of a multivariate Gaussian $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is

$$H(\mathbf{x}) \equiv -\langle \log p(\mathbf{x}) \rangle_{p(\mathbf{x})} = \frac{1}{2} \log \det(2\pi\boldsymbol{\Sigma}) + \frac{D}{2} \quad (8.6.16)$$

where $D = \dim \mathbf{x}$. Note that the entropy is independent of the mean $\boldsymbol{\mu}$.

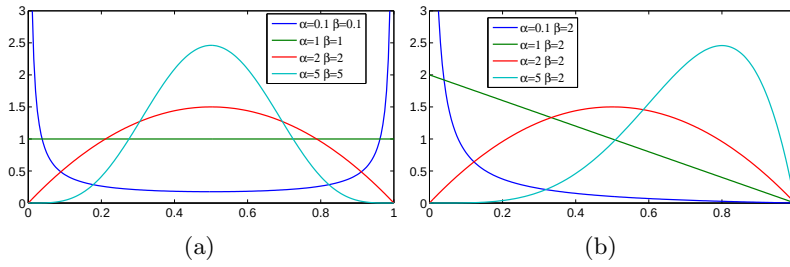


Figure 8.7: Beta distribution. The parameters α and β can also be written in terms of the mean and variance, leading to an alternative parameterisation, see exercise(94).

8.6.1 Conditioning as system reversal

For a joint distribution $p(\mathbf{x}, \mathbf{y})$, consider the conditional $p(\mathbf{x}|\mathbf{y})$. The statistics of $p(x|y)$ can be obtained using a linear system of the form

$$\mathbf{x} = \overleftarrow{\mathbf{A}}\mathbf{y} + \overleftarrow{\boldsymbol{\eta}} \quad (8.6.17)$$

where

$$\overleftarrow{\boldsymbol{\eta}} \sim \mathcal{N}(\overleftarrow{\boldsymbol{\eta}} | \overleftarrow{\boldsymbol{\mu}}, \overleftarrow{\boldsymbol{\Sigma}}) \quad (8.6.18)$$

and this reversed noise is uncorrelated with $\mathbf{y}$.

To show this, we need to make the statistics of $\mathbf{x}$ under this linear system match those given by the conditioning operation, (8.6.11). The mean of the linear system is given by

$$\boldsymbol{\mu}_x = \overleftarrow{\mathbf{A}}\boldsymbol{\mu}_y + \overleftarrow{\boldsymbol{\mu}} \quad (8.6.19)$$

and the covariances by (note that covariance of $\mathbf{y}$ remains unaffected by the system reversal)

$$\boldsymbol{\Sigma}_{xx} = \overleftarrow{\mathbf{A}}\boldsymbol{\Sigma}_{yy}\overleftarrow{\mathbf{A}}^\top + \overleftarrow{\boldsymbol{\Sigma}} \quad (8.6.20)$$

$$\boldsymbol{\Sigma}_{xy} = \overleftarrow{\mathbf{A}}\boldsymbol{\Sigma}_{yy} \quad (8.6.21)$$

From equation (8.6.21) we have

$$\overleftarrow{\mathbf{A}} = \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1} \quad (8.6.22)$$

which, using in equation (8.6.20), gives

$$\overleftarrow{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}_{xx} - \overleftarrow{\mathbf{A}}\boldsymbol{\Sigma}_{yy}\overleftarrow{\mathbf{A}}^\top = \boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx} \quad (8.6.23)$$

Using equation (8.6.19) we similarly obtain

$$\overleftarrow{\boldsymbol{\mu}} = \boldsymbol{\mu}_x - \overleftarrow{\mathbf{A}}\boldsymbol{\mu}_y = \boldsymbol{\mu}_x - \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\mu}_y \quad (8.6.24)$$

This means that we can write an explicit linear system of the form equation (8.6.17) where the parameters are given in terms of the statistics of the original system. These results are just a restatement of the conditioning results but shows how it may be interpreted as a linear system. This is useful in deriving results in inference with Linear Dynamical Systems.

8.6.2 Completing the square

A useful technique in manipulating Gaussians is completing the square. For example, the expression

$$e^{-\frac{1}{2}\mathbf{x}^\top\mathbf{A}\mathbf{x}+\mathbf{b}^\top\mathbf{x}} \quad (8.6.25)$$

can be transformed as follows. First we complete the square:

$$\frac{1}{2}\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{b}^\top\mathbf{x} = \frac{1}{2}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})^\top\mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) - \frac{1}{2}\mathbf{b}^\top\mathbf{A}^{-1}\mathbf{b} \quad (8.6.26)$$

Hence

$$e^{-\frac{1}{2}\mathbf{x}^\top\mathbf{A}\mathbf{x}+\mathbf{b}^\top\mathbf{x}} = \mathcal{N}(\mathbf{x}|\mathbf{A}^{-1}\mathbf{b}, \mathbf{A}^{-1}) \sqrt{\det(2\pi\mathbf{A}^{-1})} e^{\frac{1}{2}\mathbf{b}^\top\mathbf{A}^{-1}\mathbf{b}} \quad (8.6.27)$$

From this one can derive

$$\int e^{-\frac{1}{2}\mathbf{x}^\top\mathbf{A}\mathbf{x}+\mathbf{b}^\top\mathbf{x}} d\mathbf{x} = \sqrt{\det(2\pi\mathbf{A}^{-1})} e^{\frac{1}{2}\mathbf{b}^\top\mathbf{A}^{-1}\mathbf{b}} \quad (8.6.28)$$

8.6.3 Gaussian propagation

Let $\mathbf{y}$ be linearly related to $\mathbf{x}$ through

$$\mathbf{y} = \mathbf{M}\mathbf{x} + \boldsymbol{\eta} \quad (8.6.29)$$

where $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x)$.

Then the marginal $p(\mathbf{y}) = \int_{\mathbf{x}} p(\mathbf{y}|\mathbf{x})p(\mathbf{x})$ is a Gaussian

$$p(\mathbf{y}) = \mathcal{N}\left(\mathbf{y} | \mathbf{M}\boldsymbol{\mu}_x + \boldsymbol{\mu}, \mathbf{M}\boldsymbol{\Sigma}_x\mathbf{M}^\top + \boldsymbol{\Sigma}\right) \quad (8.6.30)$$

8.6.4 Whitening and centering

For a set of data $\mathbf{x}^1, \dots, \mathbf{x}^N$, with $\dim \mathbf{x}^n = D$, we can transform this data to $\mathbf{y}^1, \dots, \mathbf{y}^N$ with zero mean using *centering*:

$$\mathbf{y}^n = \mathbf{x}^n - \mathbf{m} \quad (8.6.31)$$

where the mean $\mathbf{m}$ of the data is given by

$$\mathbf{m} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}^n \quad (8.6.32)$$

Furthermore, we can transform to a values $\mathbf{z}^1, \dots, \mathbf{z}^N$ that have zero mean and unit covariance using *whitening*

$$\mathbf{z}^n = \mathbf{S}^{-\frac{1}{2}} (\mathbf{x}^n - \mathbf{m}) \quad (8.6.33)$$

where the covariance $\mathbf{S}$ of the data is given by

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}^n - \mathbf{m})(\mathbf{x}^n - \mathbf{m})^\top \quad (8.6.34)$$

An equivalent approach is to compute the SVD decomposition of the matrix of centered datapoints

$$\mathbf{U}\mathbf{S}\mathbf{V}^\top = \mathbf{Y}, \quad \mathbf{Y} = [\mathbf{y}^1, \dots, \mathbf{y}^N] \quad (8.6.35)$$

then

$$\mathbf{Z} = \sqrt{N} \text{diag}(1/S_{1,1}, \dots, 1/S_{D,D}) \mathbf{U}^\top \mathbf{Y} \quad (8.6.36)$$

has zero mean and unit covariance, see exercise(111).

8.6.5 Maximum likelihood training

Given a set of training data $\mathcal{X} = \{\mathbf{x}^1, \dots, \mathbf{x}^N\}$, drawn from a Gaussian $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with unknown mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, how can we find these parameters? Assuming the data are drawn i.i.d. the log likelihood is

$$L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \equiv \sum_{n=1}^N \log p(\mathbf{x}^n | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2} \sum_{n=1}^N (\mathbf{x}^n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}^n - \boldsymbol{\mu}) - \frac{N}{2} \log \det(2\pi\boldsymbol{\Sigma}) \quad (8.6.37)$$

Optimal μ

Taking the partial derivative with respect to the vector μ we obtain the vector derivative

$$\nabla_{\mu} L(\mu, \Sigma) = \sum_{n=1}^N \Sigma^{-1} (\mathbf{x}^n - \mu) \quad (8.6.38)$$

Equating to zero gives that at the optimum of the log likelihood,

$$\sum_{n=1}^N \Sigma^{-1} \mathbf{x}^n = N \mu \Sigma^{-1} \quad (8.6.39)$$

and therefore, optimally

$$\mu = \frac{1}{N} \sum_{n=1}^N \mathbf{x}^n \quad (8.6.40)$$

Optimal Σ

The derivative of L with respect to the matrix Σ requires more work. It is convenient to isolate the dependence on the covariance, and also parameterise using the inverse covariance, Σ^{-1} ,

$$L = -\frac{1}{2} \text{trace} \left(\Sigma^{-1} \underbrace{\sum_{n=1}^N (\mathbf{x}^n - \mu) (\mathbf{x}^n - \mu)^{\top}}_{\equiv \mathbf{M}} \right) + \frac{N}{2} \log \det (2\pi \Sigma^{-1}) \quad (8.6.41)$$

Using $\mathbf{M} = \mathbf{M}^{\top}$, we obtain

$$\frac{\partial}{\partial \Sigma^{-1}} L = -\frac{1}{2} \mathbf{M} + \frac{N}{2} \Sigma \quad (8.6.42)$$

Equating the derivative to the zero matrix and solving for Σ gives

$$\Sigma = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}^n - \mu) (\mathbf{x}^n - \mu)^{\top} \quad (8.6.43)$$

Equations (8.6.40) and (8.6.43) define the Maximum Likelihood solution mean and covariance for training data $\mathcal{X}$. Consistent with our previous results, in fact these equations simply set the parameters to their sample statistics of the empirical distribution. That is, the mean is set to the sample mean of the data and the covariance to the sample covariance.

8.6.6 Bayesian Inference of the mean and variance

For simplicity here we deal with the univariate case. Assuming i.i.d. data the likelihood is

$$p(\mathcal{X}|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{N/2}} e^{-\frac{1}{2\sigma^2} \sum_{n=1}^N (x^n - \mu)^2} \quad (8.6.44)$$

For a Bayesian treatment, we require the posterior of the parameters

$$p(\mu, \sigma^2|\mathcal{X}) \propto p(\mathcal{X}|\mu, \sigma^2) p(\mu, \sigma^2) = p(\mathcal{X}|\mu, \sigma^2) p(\mu|\sigma^2) p(\sigma^2) \quad (8.6.45)$$

Our aim is to find conjugate priors for the mean and variance. A convenient choice for a prior on the mean μ is that it is a Gaussian centred on μ_0 :

$$p(\mu|\mu_0, \sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2} (\mu_0 - \mu)^2} \quad (8.6.46)$$

The posterior is then

$$p(\mu, \sigma^2 | \mathcal{X}) \propto \frac{1}{\sqrt{\sigma_0^2}} \frac{1}{(\sigma^2)^{N/2}} e^{-\frac{1}{2\sigma_0^2}(\mu_0 - \mu)^2 - \frac{1}{2\sigma^2} \sum_n (x^n - \mu)^2} p(\sigma^2) \quad (8.6.47)$$

It is convenient to write this in the form

$$p(\mu, \sigma^2 | \mathcal{X}) = p(\mu | \sigma^2, \mathcal{X}) p(\sigma^2 | \mathcal{X}) \quad (8.6.48)$$

Since equation (8.6.47) has quadratic contributions in μ in the exponent, the conditional posterior $p(\mu | \sigma^2, \mathcal{X})$ is Gaussian. To identify this Gaussian we multiply out the terms in the exponent to arrive at

$$\exp -\frac{1}{2} (a\mu^2 - 2b\mu + c) \quad (8.6.49)$$

with

$$a = \frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}, \quad b = \frac{\mu_0}{\sigma_0^2} + \frac{\sum_n x^n}{\sigma^2}, \quad c = \frac{\mu_0^2}{\sigma_0^2} + \sum_n \frac{(x^n)^2}{\sigma^2} \quad (8.6.50)$$

Using the identity

$$a\mu^2 - 2b\mu + c = a \left(\mu - \frac{b}{a} \right)^2 + \left(c - \frac{b^2}{a} \right) \quad (8.6.51)$$

we can write

$$p(\mu, \sigma^2 | \mathcal{X}) \propto \underbrace{\sqrt{a} e^{-\frac{1}{2}a(\mu - \frac{b}{a})^2}}_{p(\mu | \mathcal{X}, \sigma^2)} \underbrace{\frac{1}{\sqrt{a}} e^{-\frac{1}{2}(c - \frac{b^2}{a})} \frac{1}{\sqrt{\sigma_0^2}} \frac{1}{(\sigma^2)^{N/2}} p(\sigma^2)}_{p(\sigma^2 | \mathcal{X})} \quad (8.6.52)$$

We encounter a difficulty in attempting to find a conjugate prior for σ^2 because the term b^2/a is not a simple expression of σ^2 . For this reason we constrain

$$\sigma_0^2 \equiv \gamma \sigma^2 \quad (8.6.53)$$

for some fixed hyperparameter γ . Defining the constants

$$\tilde{a} = \frac{1}{\gamma} + N, \quad \tilde{b} = \frac{\mu_0}{\gamma} + \sum_n x^n, \quad \tilde{c} = \frac{\mu_0^2}{\gamma} + \sum_n (x^n)^2 \quad (8.6.54)$$

we have

$$c - \frac{b^2}{a} = \frac{1}{\sigma^2} \left(\tilde{c} - \frac{\tilde{b}^2}{\tilde{a}} \right) \quad (8.6.55)$$

Using this expression in equation (8.6.52) we obtain

$$p(\sigma^2 | \mathcal{X}) \propto (\sigma^2)^{-N/2} e^{-\frac{1}{2\sigma^2} \left(\tilde{c} - \frac{\tilde{b}^2}{\tilde{a}} \right)} p(\sigma^2) \quad (8.6.56)$$

If we therefore use an inverse gamma distribution we will have a conjugate prior for σ^2 . For a Gauss-Inverse-Gamma prior:

$$p(\mu, \sigma^2) = \mathcal{N}(\mu | \mu_0, \gamma \sigma^2) \text{InvGam}(\sigma^2 | \alpha, \beta) \quad (8.6.57)$$

the posterior is also *Gauss-Inverse-Gamma* with

$$p(\mu, \sigma^2 | \mathcal{X}) = \mathcal{N} \left(\mu \left| \frac{\tilde{b}}{\tilde{a}}, \frac{\sigma^2}{\tilde{a}} \right. \right) \text{InvGam} \left(\sigma^2 \left| \alpha + \frac{N}{2}, \beta + \frac{1}{2} \left(\tilde{c} - \frac{\tilde{b}^2}{\tilde{a}} \right) \right. \right) \quad (8.6.58)$$

8.6.7 Gauss-Gamma distribution

It is common to use a prior on the *precision*, defined as the inverse variance

$$\lambda \equiv \frac{1}{\sigma^2} \quad (8.6.59)$$

If we then use a Gamma prior

$$p(\lambda|\alpha, \beta) = \text{Gam}(\lambda|\alpha, \beta) = \frac{1}{\beta^\alpha \Gamma(\alpha)} \lambda^{\alpha-1} e^{-\lambda/\beta} \quad (8.6.60)$$

The posterior will be

$$p(\lambda|\mathcal{X}, \alpha, \beta) = \text{Gam}(\lambda|\alpha + N/2, \tilde{\beta}) \quad (8.6.61)$$

where

$$\frac{1}{\tilde{\beta}} = \frac{1}{\beta} + \frac{1}{2} \left(\tilde{c} - \frac{\tilde{b}^2}{\tilde{a}} \right) \quad (8.6.62)$$

The *Gauss-Gamma* prior distribution

$$p(\mu, \lambda|\mu_0, \alpha, \beta, \gamma) = \mathcal{N}(\mu|\mu_0, \gamma\lambda^{-1}) \text{Gam}(\lambda|\alpha, \beta) \quad (8.6.63)$$

is the conjugate prior for a Gaussian with unknown mean μ and precision λ .

The posterior for this prior is a Gauss-Gamma distribution with parameters

$$p(\mu, \lambda|\mathcal{X}, \mu_0, \alpha, \beta, \gamma) = \mathcal{N}\left(\mu \middle| \frac{\tilde{b}}{\tilde{a}}, \frac{1}{\tilde{a}\lambda}\right) \text{Gam}(\lambda|\alpha + N/2, \tilde{\beta}) \quad (8.6.64)$$

The marginal $p(\mu|\mathcal{X}, \mu_0, \alpha, \beta, \gamma)$ is a Student's t -distribution. An example of a Gauss-Gamma prior/posterior is given in fig(8.8).

The Maximum Likelihood solution is recovered in the limit of a ‘flat’ (improper) prior $\mu_0 = 0, \gamma \rightarrow \infty, \alpha = 1/2, \beta \rightarrow \infty$, see exercise(102). The unbiased estimators for the mean and variance are given using the proper prior $\mu_0 = 0, \gamma \rightarrow \infty, \alpha = 1, \beta \rightarrow \infty$, exercise(103).

For the multivariate case, the extension of these techniques uses a multivariate Gaussian distribution for the conjugate prior on the mean, and an Inverse Wishart distribution for the conjugate prior on the covariance[124].

8.7 Exponential Family

A theoretically convenient class of distributions are the exponential family, which contains many standard distributions, including the Gaussian, Gamma, Poisson, Dirichlet, Wishart, Multinomial, Markov Random Field.

Definition 82 (Exponential Family). For a distribution on a (possibly multidimensional) variable x (continuous or discrete) an *exponential family* model is of the form

$$p(x|\boldsymbol{\theta}) = h(x) e^{\sum_i \eta_i(\boldsymbol{\theta}) T_i(x) - \psi(\boldsymbol{\theta})} \quad (8.7.1)$$

$\boldsymbol{\theta}$ are the parameters, $T_i(x)$ the test statistics, and $\psi(\boldsymbol{\theta})$ is the log partition function that ensure normalisation

$$\psi(\boldsymbol{\theta}) = \log \int_x h(x) e^{\sum_i \eta_i(\boldsymbol{\theta}) T_i(x)} \quad (8.7.2)$$

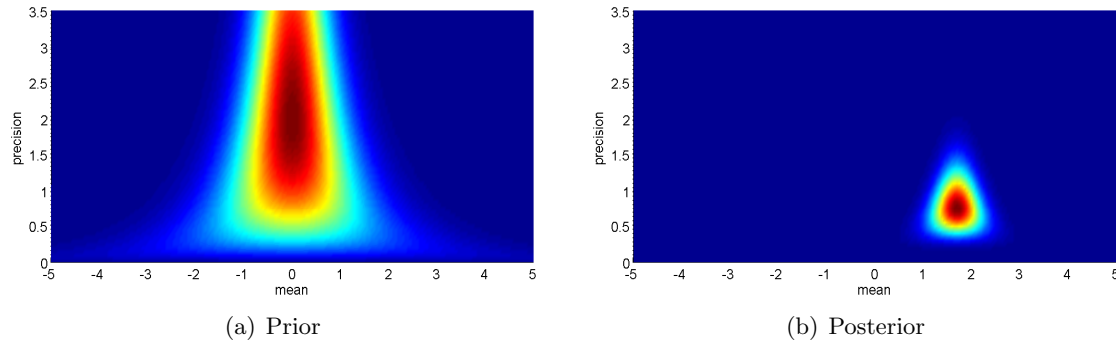


Figure 8.8: Bayesian approach to inferring the mean and precision (inverse variance) of a Gaussian based on $N = 10$ randomly drawn datapoints. **(a)**: A Gauss-Gamma prior with $\mu_0 = 0$, $\alpha = 2$, $\beta = 1$, $\gamma = 1$. **(b)**: Gauss-Gamma posterior conditional on the data. For comparison, the sample mean of the data is 1.87 and Maximum Likelihood optimal variance is 1.16 (computed using the N normalisation). The 10 datapoints were drawn from a Gaussian with mean 2 and variance 1. See `demoGaussBayes.m`.

One can always transform the parameters to the form $\boldsymbol{\eta}(\boldsymbol{\theta}) = \boldsymbol{\theta}$ in which case the distribution is in *canonical form*:

$$p(x|\boldsymbol{\theta}) = h(x)e^{\boldsymbol{\theta}^T \mathbf{T}(x) - \psi(\boldsymbol{\theta})} \quad (8.7.3)$$

For example the univariate Gaussian can be written

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} = e^{-\frac{1}{2\sigma^2}x^2 + \frac{\mu}{\sigma^2}x - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log \pi\sigma^2} \quad (8.7.4)$$

Defining $t_1(x) = x$, $t_2(x) = -x^2/2$ and $\theta_1 = \mu$, $\theta_2 = \sigma^2$, $h(x) = 1$, then

$$\eta_1(\boldsymbol{\theta}) = \frac{\theta_1}{\theta_2}, \quad \eta_2(\boldsymbol{\theta}) = \frac{1}{\theta_2}, \quad \psi(\boldsymbol{\theta}) = \frac{1}{2} \left(\frac{\theta_1^2}{\theta_2} + \log \pi\theta_2 \right) \quad (8.7.5)$$

Note that the parameterisation is not necessarily unique – we can for example rescale the functions $T_i(x)$ and inversely scale η_i by the same amount to arrive at an equivalent representation.

8.7.1 Conjugate priors

In principle, Bayesian learning for the exponential family is straightforward. In canonical form

$$p(x|\boldsymbol{\theta}) = h(x)e^{\boldsymbol{\theta}^T \mathbf{T}(x) - \psi(\boldsymbol{\theta})} \quad (8.7.6)$$

For a prior with hyperparameters $\boldsymbol{\alpha}, \gamma$,

$$p(\boldsymbol{\theta}|\boldsymbol{\alpha}, \gamma) \propto e^{\boldsymbol{\theta}^T \boldsymbol{\alpha} - \gamma\psi(\boldsymbol{\theta})} \quad (8.7.7)$$

the posterior is

$$p(\boldsymbol{\theta}|x) \propto p(x|\boldsymbol{\theta})p(\boldsymbol{\theta}) \quad (8.7.8)$$

$$\propto h(x)e^{\boldsymbol{\theta}^T \mathbf{T}(x) - \psi(\boldsymbol{\theta})} e^{\boldsymbol{\theta}^T \boldsymbol{\alpha} - \gamma\psi(\boldsymbol{\theta})} \quad (8.7.9)$$

$$\propto e^{\boldsymbol{\theta}^T [\mathbf{T}(x) + \boldsymbol{\alpha}] - [\gamma+1]\psi(\boldsymbol{\theta})} \quad (8.7.10)$$

so that the prior equation (8.7.7) is conjugate for the exponential family likelihood equation (8.7.6). Whilst the likelihood is in the exponential family, the conjugate prior is not necessarily in the exponential family.

8.8 The Kullback-Leibler Divergence $KL(q|p)$

The Kullback-Leibler divergence $KL(q|p)$ measures the ‘difference’ between distributions q and p [68].

Definition 83. KL divergence For two distributions $q(x)$ and $p(x)$

$$KL(q|p) \equiv \langle \log q(x) - \log p(x) \rangle_{q(x)} \geq 0 \quad (8.8.1)$$

where $\langle f(x) \rangle_{r(x)}$ denotes average of the function $f(x)$ with respect to the distribution $r(x)$.

The KL divergence is ≥ 0

The KL divergence is widely used and it is therefore important to understand why the divergence is positive.

To see this, consider the following linear bound on the function $\log(x)$

$$\log(x) \leq x - 1 \quad (8.8.2)$$

as plotted in the figure on the right. Replacing x by $p(x)/q(x)$ in the above bound

$$\frac{p(x)}{q(x)} - 1 \geq \log \frac{p(x)}{q(x)} \quad (8.8.3)$$

Since probabilities are non-negative, we can multiply both sides by $q(x)$ to obtain

$$p(x) - q(x) \geq q(x) \log p(x) - q(x) \log q(x) \quad (8.8.4)$$

We now integrate (or sum in the case of discrete variables) both sides. Using $\int p(x)dx = 1$, $\int q(x)dx = 1$,

$$1 - 1 \geq \langle \log p(x) - \log q(x) \rangle_{q(x)} \quad (8.8.5)$$

Rearranging gives

$$\langle \log q(x) - \log p(x) \rangle_{q(x)} \equiv KL(q|p) \geq 0 \quad (8.8.6)$$

The KL divergence is zero if and only if the two distributions are exactly the same.

8.8.1 Entropy

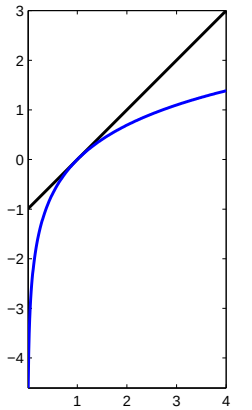
For both discrete and continuous variables, the *entropy* is defined as

$$H(p) \equiv -\langle \log p(x) \rangle_{p(x)} \quad (8.8.7)$$

For continuous variables, this is also called the *differential entropy*, see also exercise(113). The entropy is a measure of the uncertainty in a distribution. One way to see this is that

$$H(p) = -KL(p|u) + \text{const.} \quad (8.8.8)$$

where u is a uniform distribution. Since the $KL(p|u) \geq 0$, the less like a uniform distribution p is, the smaller will be the entropy. Or, vice versa, the more similar p is to a uniform distribution, the greater will be the entropy. Since the uniform distribution contains the least information a priori about which state $p(x)$ is in, the entropy is therefore a measure of the a priori uncertainty in the state occupancy. For a discrete distribution we can permute the state labels without changing the entropy. For a discrete distribution the entropy is positive, whereas the differential entropy can be negative.



8.9 Code

`demoGaussBayes.m`: Bayesian fitting of a univariate Gaussian

`logGaussGamma.m`: Plotting routine for a Gauss-Gamma distribution

8.10 Exercises

Exercise 80. In a public lecture, the following phrase was uttered by a Professor of Experimental Psychology: ‘In a recent data survey, 90% of people claim to have above average intelligence, which is clearly nonsense!’ [Audience Laughs]. Is it theoretically possible for 90% of people to have above average intelligence? If so, give an example, otherwise explain why not. What about above median intelligence?

Exercise 81. Consider the distribution defined on real variables x, y :

$$p(x, y) \propto (x^2 + y^2)^2 e^{-x^2 - y^2}, \quad \text{dom}(x) = \text{dom}(y) = \{-\infty \dots \infty\} \quad (8.10.1)$$

Show that $\langle x \rangle = \langle y \rangle = 0$. Furthermore show that x and y are uncorrelated, $\langle xy \rangle = \langle x \rangle \langle y \rangle$. Whilst x and y are uncorrelated, show that they are nevertheless dependent.

Exercise 82. For a variable x with $\text{dom}(x) = \{0, 1\}$, and $p(x = 1) = \theta$, show that in n independent draws $x_1, \dots, x_n$ from this distribution, the probability of observing k states 1 is the Binomial distribution

$$\binom{n}{k} \theta^k (1 - \theta)^{n-k} \quad (8.10.2)$$

Exercise 83 (Normalisation constant of a Gaussian). The normalisation constant of a Gaussian distribution is related to the integral

$$I = \int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2} dx \quad (8.10.3)$$

By considering

$$I^2 = \int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2} dx \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2 + y^2} dx dy \quad (8.10.4)$$

and transforming to polar coordinates, show that

1. $I = \sqrt{2\pi}$
2. $\int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx = \sqrt{2\pi\sigma^2}$

Exercise 84. For a univariate Gaussian distribution, show that

1. $\mu = \langle x \rangle_{\mathcal{N}(x|\mu, \sigma^2)}$
2. $\sigma^2 = \langle (x - \mu)^2 \rangle_{\mathcal{N}(x|\mu, \sigma^2)}$

Exercise 85. Show that the marginal of a Dirichlet distribution is another Dirichlet distribution:

$$\int_{\theta_j} \text{Dirichlet}(\theta|\mathbf{u}) = \text{Dirichlet}(\theta_{\setminus j}|\mathbf{u}_{\setminus j}) \quad (8.10.5)$$

Exercise 86. For a Beta distribution, show that

$$\langle x^k \rangle = \frac{B(\alpha + k, \beta)}{B(\alpha, \beta)} \quad (8.10.6)$$

and, using $\Gamma(x + 1) = x\Gamma(x)$, derive an explicit expression for the k^{th} moment of a Beta distribution.

Exercise 87. Define the **moment generating function** as

$$g(t) \equiv \langle e^{tx} \rangle_{p(x)} \quad (8.10.7)$$

Show that

$$\lim_{t \rightarrow 0} \frac{d^k}{dt^k} g(t) = \langle x^k \rangle_{p(x)} \quad (8.10.8)$$

Exercise 88 (Change of variables). Consider a one dimensional continuous random variable x with corresponding $p(x)$. For a variable $y = f(x)$, where $f(x)$ is a monotonic function, show that the distribution of y is

$$p(y) = p(x) \left(\frac{df}{dx} \right)^{-1}, \quad x = f^{-1}(y) \quad (8.10.9)$$

Exercise 89 (Normalisation of a Multivariate Gaussian). Consider

$$I = \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} dx \quad (8.10.10)$$

By using the transformation

$$z = \Sigma^{-\frac{1}{2}}(x - \mu) \quad (8.10.11)$$

show that

$$I = \sqrt{\det(2\pi\Sigma)} \quad (8.10.12)$$

Exercise 90. Consider the partitioned matrix

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \quad (8.10.13)$$

for which we wish to find the inverse $\mathbf{M}^{-1}$. We assume that $\mathbf{A}$ is $m \times m$ and invertible, and $\mathbf{D}$ is $n \times n$ and invertible. By definition, the partitioned inverse

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{P} & \mathbf{Q} \\ \mathbf{R} & \mathbf{S} \end{pmatrix} \quad (8.10.14)$$

must satisfy

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{P} & \mathbf{Q} \\ \mathbf{R} & \mathbf{S} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_m & 0 \\ 0 & \mathbf{I}_n \end{pmatrix} \quad (8.10.15)$$

where in the above $\mathbf{I}_m$ is the $m \times m$ identity matrix of the same dimension as $\mathbf{A}$, and 0 the zero matrix of the same dimension as $\mathbf{D}$. Using the above, derive the results

$$\begin{aligned} \mathbf{P} &= (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} \\ \mathbf{Q} &= -\mathbf{A}^{-1}\mathbf{B}(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \\ \mathbf{R} &= -\mathbf{D}^{-1}\mathbf{C}(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} \\ \mathbf{S} &= (\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \end{aligned}$$

Exercise 91. Show that for Gaussian distribution $p(x) = \mathcal{N}(x|\mu, \sigma^2)$ the skewness and kurtosis are both zero.

Exercise 92. Consider a small interval of time δt and let the probability of an event occurring in this small interval be $\theta \delta t$. Derive a distribution that expresses the probability of at least one event in an interval from 0 to t .

Exercise 93. Consider a vector variable $\mathbf{x} = (x_1, \dots, x_n)$ and set of functions defined on each component of x , $\phi_i(x_i)$. For example for $\mathbf{x} = (x_1, x_2)$ we might have

$$\phi_1(x_1) = -|x_1|, \quad \phi_2(x_2) = -x_2^2 \quad (8.10.16)$$

Consider the distribution

$$p(\mathbf{x}|\theta) = \frac{1}{Z} e^{\theta^T \phi(\mathbf{x})} \quad (8.10.17)$$

where $\phi(\mathbf{x})$ is a vector function with i^{th} component $\phi_i(x_i)$, and θ is a parameter vector. Each component is tractably integrable in the sense that

$$\int_{-\infty}^{\infty} e^{\theta_i \phi_i(x_i)} dx_i \quad (8.10.18)$$

can be computed either analytically or to an acceptable numerical accuracy. Show that

1. $x_i \perp\!\!\!\perp x_j | \emptyset$.
2. The normalisation constant Z can be tractably computed.
3. Consider the transformation

$$\mathbf{x} = \mathbf{M}\mathbf{y} \quad (8.10.19)$$

for an invertible matrix $\mathbf{M}$. Show that the distribution $p(\mathbf{y}|\mathbf{M}, \theta)$ is tractable (its normalisation constant is known), and that, in general, $y_i \perp\!\!\!\perp y_j | \emptyset$. Explain the significance of this in deriving tractable multivariate distributions.

Exercise 94. Show that we may reparameterise the Beta distribution, definition(72) by writing the parameters α and β as functions of the mean m and variance s using

$$\alpha = \beta\gamma, \quad \gamma \equiv m/(1-m) \quad (8.10.20)$$

$$\beta = \frac{1}{1+\gamma} \left(\frac{\gamma}{s(1+\gamma)^2} - 1 \right) \quad (8.10.21)$$

Exercise 95. Consider the function

$$f(\gamma + \alpha, \beta, \theta) \equiv \theta^{\gamma+\alpha-1} (1-\theta)^{\beta-1} \quad (8.10.22)$$

show that

$$\lim_{\gamma \rightarrow 0} \frac{\partial}{\partial \gamma} f(\gamma + \alpha, \beta, \theta) = \theta^{\alpha-1} (1-\theta)^{\beta-1} \log \theta \quad (8.10.23)$$

and hence that

$$\int \theta^{\alpha-1} (1-\theta)^{\beta-1} \log \theta d\theta = \lim_{\gamma \rightarrow 0} \frac{\partial}{\partial \gamma} \int f(\gamma + \alpha, \beta, \theta) d\theta = \frac{\partial}{\partial \alpha} \int f(\alpha, \beta, \theta) d\theta \quad (8.10.24)$$

Using this result, show therefore that

$$\langle \log \theta \rangle_{B(\theta|\alpha, \beta)} = \frac{\partial}{\partial \alpha} \log B(\alpha, \beta) \quad (8.10.25)$$

where $B(\alpha, \beta)$ is the Beta function. Show additionally that

$$\langle \log (1-\theta) \rangle_{B(\theta|\alpha, \beta)} = \frac{\partial}{\partial \beta} \log B(\alpha, \beta) \quad (8.10.26)$$

Using the fact that

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)} \quad (8.10.27)$$

where $\Gamma(x)$ is the gamma function, relate the above averages to the **digamma function**, defined as

$$\psi(x) = \frac{d}{dx} \log \Gamma(x) \quad (8.10.28)$$

Exercise 96. Using a similar ‘generating function’ approach as in exercise(95), explain how to compute

$$\langle \log \theta_i \rangle_{\text{Dirichlet}(\theta|\mathbf{u})} \quad (8.10.29)$$

Exercise 97. Consider the function

$$f(x) = \int_0^\infty \delta \left(\sum_{i=1}^n \theta_i - x \right) \prod_i \theta_i^{u_i-1} d\theta_1 \dots d\theta_n \quad (8.10.30)$$

Show that the Laplace transform of $f(x)$, $\tilde{f}(s) \equiv \int_0^\infty e^{-sx} f(x) dx$ is

$$\tilde{f}(s) = \prod_{i=1}^n \left\{ \int_0^\infty e^{-s\theta_i} \theta_i^{u_i-1} d\theta_i \right\} = \frac{1}{s^{\sum_i u_i}} \prod_{i=1}^n \Gamma(u_i) \quad (8.10.31)$$

By taking the inverse Laplace transform, show that

$$f(x) = \frac{\prod_{i=1}^n \Gamma(u_i)}{\Gamma(\sum_i u_i - 1)} x^{\sum_i u_i - 1} \quad (8.10.32)$$

Hence show that the normalisation constant of a Dirichlet distribution with parameters $\mathbf{u}$ is given by

$$\frac{\prod_{i=1}^n \Gamma(u_i)}{\Gamma(\sum_i u_i)} \quad (8.10.33)$$

Exercise 98. By using the Laplace transform, as in exercise(97), show that the marginal of a Dirichlet distribution is a Dirichlet distribution.

Exercise 99. Derive the formula for the differential entropy of a multi-variate Gaussian.

Exercise 100. Show that for a gamma distribution $\text{Gam}(x|\alpha, \beta)$ the mode is given by

$$x^* = (\alpha - 1) \beta \quad (8.10.34)$$

provided that $\alpha \geq 1$.

Exercise 101. Consider a distribution $p(x|\theta)$ and a distribution with θ changed by a small amount, δ . Take the Taylor expansion of

$$KL(p(x|\theta)|p(x|\theta + \delta)) \quad (8.10.35)$$

for small δ and show that this is equal to

$$-\frac{\delta^2}{2} \left\langle \frac{\partial^2}{\partial \theta^2} \log p(x|\theta) \right\rangle_{p(\theta)} \quad (8.10.36)$$

More generally for a distribution parameterised by a vector $\theta_i + \delta_i$, show that a small change in the parameter results in

$$\sum_{i,j} \frac{\delta_i \delta_j}{2} F_{ij} \quad (8.10.37)$$

where the **Fisher Information** matrix is defined as

$$F_{ij} = - \left\langle \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log p(x|\theta) \right\rangle_{p(\theta)} \quad (8.10.38)$$

Show that the Fisher information matrix is positive (semi) definite by expressing it equivalently as

$$F_{ij} = \left\langle \frac{\partial}{\partial \theta_i} \log p(x|\theta) \frac{\partial}{\partial \theta_j} \log p(x|\theta) \right\rangle_{p(\theta)} \quad (8.10.39)$$

Exercise 102. Consider the joint prior distribution

$$p(\mu, \lambda | \mu_0, \alpha, \beta, \gamma) = \mathcal{N}(\mu | \mu_0, \gamma \lambda^{-1}) \text{Gam}(\lambda | \alpha, \beta) \quad (8.10.40)$$

Show that for $\mu_0 = 0$, $\gamma \rightarrow \infty$, $\beta \rightarrow \infty$, then the prior distribution becomes ‘flat’ (independent of μ and λ) for $\alpha = 1/2$. Show that for these settings the mean and variance that jointly maximise the posterior equation (8.6.64) are given by the standard Maximum Likelihood settings

$$\mu_* = \frac{1}{N} \sum_n x^n, \quad \sigma_*^2 = \frac{1}{N} \sum_n (x^n - \mu_*)^2 \quad (8.10.41)$$

Exercise 103. Show that in the limit $\mu_0 = 0, \gamma \rightarrow \infty, \alpha = 1, \beta \rightarrow \infty$, the jointly optimal mean and variance obtained from

$$\operatorname{argmax}_{\mu, \lambda} p(\mu, \lambda | \mathcal{X}, \alpha, \beta, \gamma) \quad (8.10.42)$$

is given by

$$\mu_* = \frac{1}{N} \sum_n x^n, \quad \sigma_*^2 = \frac{1}{N+1} \sum_n (x^n - \mu_*)^2 \quad (8.10.43)$$

where $\sigma_*^2 = 1/\lambda_*$. Note that these correspond to the standard ‘unbiased’ estimators of the mean and variance.

Exercise 104. For the Gauss-Gamma posterior $p(\mu, \lambda | \mu_0, \alpha, \beta, \mathcal{X})$ given in equation (8.6.64) compute the marginal posterior $p(\mu | \mu_0, \alpha, \beta, \mathcal{X})$. What is the mean of this distribution?

Exercise 105. Derive equation (8.6.30).

Exercise 106. Consider the multivariate Gaussian distribution $p(\mathbf{x}) \sim \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma})$ on the vector $\mathbf{x}$ with components $x_1, \dots, x_n$:

$$p(\mathbf{x}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})} \quad (8.10.44)$$

Calculate $p(x_i | x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$.

Exercise 107. Observations $y_0, \dots, y_{n-1}$ are noisy i.i.d. measurements of an underlying variable x with $p(x) \sim \mathcal{N}(x | 0, \sigma_0^2)$ and $p(y_i | x) \sim \mathcal{N}(y_i | x, \sigma^2)$ for $i = 0, \dots, n-1$. Show that $p(x | y_0, \dots, y_{n-1})$ is Gaussian with mean

$$\mu = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \bar{y} \quad (8.10.45)$$

where $\bar{y} = (y_0 + y_1 + \dots + y_{n-1})/n$ and variance σ_n^2 such that

$$\frac{1}{\sigma_n^2} = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}. \quad (8.10.46)$$

Exercise 108. Consider a set of data $x^1, \dots, x^N$ drawn from a Gaussian with known mean μ and unknown variance σ^2 . Assume a gamma distribution prior on $\tau = 1/\sigma^2$,

$$p(\tau) = \text{Gam}^{is}(\tau | a, b) \quad (8.10.47)$$

1. Show that the posterior distribution is

$$p(\tau | \mathcal{X}) = \text{Gam}^{is} \left(\tau | a + \frac{N}{2}, b + \frac{1}{2} \sum_{n=1}^N (x^n - \mu)^2 \right) \quad (8.10.48)$$

2. Show that the distribution for x is

$$p(x|\mathcal{X}) = \int p(x|\tau)p(\tau|\mathcal{X})d\tau = \text{Student}\left(x|\mu, \lambda = \frac{a}{b}, \nu = 2a\right) \quad (8.10.49)$$

Exercise 109. The Poisson distribution is a discrete distribution on the non-negative integers, with

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad x = 0, 1, 2, \dots \quad (8.10.50)$$

You are given a sample of n observations $x_1, \dots, x_n$ drawn from this distribution. Determine the maximum likelihood estimator of the Poisson parameter λ .

Exercise 110. For a Gaussian mixture model

$$p(\mathbf{x}) = \sum_i p_i \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), \quad p_i > 0, \sum_i p_i = 1 \quad (8.10.51)$$

show that $p(\mathbf{x})$ has mean

$$\langle \mathbf{x} \rangle = \sum_i p_i \boldsymbol{\mu}_i \quad (8.10.52)$$

and covariance

$$\sum_i p_i \left(\boldsymbol{\Sigma}_i + \boldsymbol{\mu}_i \boldsymbol{\mu}_i^T \right) - \sum_i p_i \boldsymbol{\mu}_i \sum_j p_j \boldsymbol{\mu}_j^T \quad (8.10.53)$$

Exercise 111. Show that for the whitened data matrix $\mathbf{x}$, given in equation (8.6.36), $\mathbf{Z}\mathbf{Z}^T = N\mathbf{I}$.

Exercise 112. Consider a uniform distribution $p_i = 1/N$ defined on states $i = 1, \dots, N$. Show that the entropy of this distribution is

$$H = - \sum_{i=1}^N p_i \log p_i = \log N \quad (8.10.54)$$

and that there for as the number of states N increases to infinity, the entropy diverges to infinity.

Exercise 113. Consider a continuous distribution $p(x)$, $x \in [0, 1]$. We can form a discrete approximation with probabilities p_i to this continuous distribution by identifying a continuous value i/N for each state $i = 1, \dots, N$. With this

$$p_i = \frac{p(i/N)}{\sum_i p(i/N)} \quad (8.10.55)$$

show that the entropy $H = - \sum_i p_i \log p_i$ is given by

$$H = - \frac{1}{\sum_i p(i/N)} \sum_i p(i/N) \log p(i/N) + \log \sum_i p(i/N) \quad (8.10.56)$$

Since for a continuous distribution

$$\int_0^1 p(x) dx = 1 \quad (8.10.57)$$

a discrete approximation of this integral into bins of size $1/N$ gives

$$\frac{1}{N} \sum_{i=1}^N p(i/N) = 1 \quad (8.10.58)$$

Hence show that for large N ,

$$H \approx - \int_0^1 p(x) \log p(x) dx + \text{const.} \quad (8.10.59)$$

where the constant tends to infinity as $N \rightarrow \infty$. Note that this result says that as a continuous distribution has essentially an infinite number of states, the amount of uncertainty in the distribution is infinite (alternatively, we would need an infinite number of bits to specify a continuous value). This motivates the definition of the differential entropy, which neglects the infinite constant of the limiting case of the discrete entropy.

Exercise 114. Consider two multivariate Gaussians $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$.

1. Show that the log product of the two Gaussians is given by

$$-\frac{1}{2} \mathbf{x}^T (\boldsymbol{\Sigma}_1^{-1} + \boldsymbol{\Sigma}_2^{-1}) \mathbf{x} + \mathbf{x}^T (\boldsymbol{\Sigma}_1^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_2^{-1} \boldsymbol{\mu}_2) - \frac{1}{2} (\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}_1^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2^T \boldsymbol{\Sigma}_2^{-1} \boldsymbol{\mu}_2) - \frac{1}{2} \log \det (2\pi \boldsymbol{\Sigma}_1) \det (2\pi \boldsymbol{\Sigma}_2)$$

2. Defining $\mathbf{A} = \boldsymbol{\Sigma}_1^{-1} + \boldsymbol{\Sigma}_2^{-1}$ and $\mathbf{b} = \boldsymbol{\Sigma}_1^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_2^{-1} \boldsymbol{\mu}_2$ we can write the above as

$$-\frac{1}{2} (\mathbf{x} - \mathbf{A}^{-1} \mathbf{b})^T \mathbf{A} (\mathbf{x} - \mathbf{A}^{-1} \mathbf{b}) + \frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} - \frac{1}{2} (\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}_1^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2^T \boldsymbol{\Sigma}_2^{-1} \boldsymbol{\mu}_2) - \frac{1}{2} \log \det (2\pi \boldsymbol{\Sigma}_1) \det (2\pi \boldsymbol{\Sigma}_2)$$

Writing $\boldsymbol{\Sigma} = \mathbf{A}^{-1}$ and $\boldsymbol{\mu} = \mathbf{A}^{-1} \mathbf{b}$ show that the product of Gaussians is a Gaussian with covariance

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_1 (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1} \boldsymbol{\Sigma}_2 \quad (8.10.60)$$

mean

$$\boldsymbol{\mu} = \boldsymbol{\Sigma}_1 (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1} \boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_2 (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1} \boldsymbol{\mu}_1 \quad (8.10.61)$$

and log prefactor

$$\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} - \frac{1}{2} (\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}_1^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2^T \boldsymbol{\Sigma}_2^{-1} \boldsymbol{\mu}_2) - \frac{1}{2} \log \det (2\pi \boldsymbol{\Sigma}_1) \det (2\pi \boldsymbol{\Sigma}_2) + \frac{1}{2} \log \det (2\pi \boldsymbol{\Sigma})$$

3. Show that this can be written as

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \frac{\exp \left(-\frac{1}{2} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathbf{S}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \right)}{\sqrt{\det (2\pi \mathbf{S})}} \quad (8.10.62)$$

Exercise 115. Show that

$$\frac{\partial}{\partial \theta} \langle \log p(x|\theta) \rangle_{p(x|\theta^0)} \big|_{\theta=\theta^0} = 0 \quad (8.10.63)$$